MALAWI NATIONAL EXAMINATIONS BOARD

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION (MSCE) EXAMINATION

PHYSICAL SCIENCE - PAPER I (PHYSICS)

Subject Code: M091/I | Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer ALL questions in Section A (60 Marks).
2. Answer any THREE questions in Section B (40 Marks).
3. Acceleration due to gravity g = 9.8 m/s^2.

SECTION A: MECHANICS & THERMODYNAMICS

Question 1: Newton’s Laws & Momentum

(a) State Newton’s second law of motion in terms of momentum. [2 marks]

(b) A bullet of mass 20g travelling at 300 m/s embeds into a 2kg block.
Calculate the common velocity after impact. [4 marks]

Question 2: Optics and Wave Motion

(a) Define total internal reflection and critical angle. [3 marks]

(b) A ray of light passes from glass (n = 1.5) into water (n = 1.33).
Find the critical angle at the interface. [3 marks]